Design And Analysis of Algorithms
Lecture 18: Chernoff-Hoeffding Bounds

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In this lecture, we present a number of useful tools for describing how the sum of n independent random variables $X_1, \ldots, X_n$ concentrates. Let $S_n = \sum_i X_i$ be the random variable representing the sum of n random variables: by linearity of expectation we know that $\mathbb{E}[S_n] = \sum_i \mathbb{E}[X_i]$. The question we tackles is the probability with which we observe S_n being far from its mean.

1 Central Iimit Theorem

JM: fill in

2 Markov and Chebychev

We begin by reminding ourselves about the most basic concentration inequality: Markov's inequality, which we introduced in the first lecture.

Theorem 1 (Markov's theorem)**.** *Let S_i be a nonnegative random variable. Then, for any $t > 0$,*

$$\Pr[S_i \geq t] \leq \frac{\mathbb{E}[S_i]}{t}.$$

Proof. We calculate the expected value $\mathbb{E}[S_i]$:

$$\begin{aligned}
\mathbb{E}[S_i] &= \int_{x \geq 0} x f(x) dx \\
&\geq \int_{x \geq t} x f(x) dx \\
&\geq t \int_{x \geq t} f(x) dx \\
&\geq t \mathbb{P}[S_i \geq t]
\end{aligned}$$

where the first inequality follows from nonnegativity. □

This is the starting point for proving the next concentration guarantee.

Theorem 2. *Let X be a random variable with mean μ and variance σ^2. Then*

$$\mathbb{P}[|X - \mu| \geq \lambda] \leq \frac{\sigma^2}{\lambda^2}.$$

Proof. Consider $Y = (X - \mu)^2$, then apply Markov:

$$\begin{aligned}\mathbb{P}[|X - \mu| \geq \lambda] &= \mathbb{P}[Y \geq \lambda^2] \\ &\leq \frac{\mathbb{E}[Y]}{\lambda^2} \\ &= \frac{\sigma^2}{\lambda^2}.\end{aligned}$$

□

3 Chernoff-Hoeffding bounds

Chebyschev's inequality suggests that we should be able to get tighter concentration about a random variable's mean than Markov might give us: how much further should we expect to guarantee concentration from independent random variables? The next theorem we present gives us a yet tighter

Theorem 3 (Chernoff-Hoeffding bound)**.** *Let $X_i \sim D_i$ independently for each i, bounded between 0 and 1. Then, for $S_n = \sum_i X_i$, with $\mathbb{E}[S_n] = \mu$, for any $\epsilon \geq 0$, we have*

$$\mathbb{P}[S_n \geq (1+\epsilon)\mu] \leq e^{-\frac{\mu\epsilon^2}{2+\epsilon}}$$

and

$$\mathbb{P}[S_n \leq (1-\epsilon)\mu] \leq e^{-\frac{\mu\epsilon^2}{3}}.$$

Proof. We start by assuming X_i is Bernoulli with mean μ_i; at the end we will remove this assumption. Then, X_i only takes on values 0 and 1. Consider upper-bounding the probability that S_n is small; then, for any $t > 0$, we have

$$\begin{aligned}\mathbb{P}[S_n \geq (1+\epsilon)\mu] &= \mathbb{P}[e^{tS_n} \geq e^{t(1+\epsilon)\mu}] \\ &\leq \frac{\mathbb{E}[e^{tS_n}]}{e^{t(1+\epsilon)\mu}} && \text{Markov's Inequality} \\ &\leq \frac{\mathbb{E}[\prod_i e^{tX_i}]}{e^{t(1+\epsilon)\mu}} \\ &= \frac{\prod_i \mathbb{E}[e^{tX_i}]}{e^{t(1+\epsilon)\mu}} && \text{Independence of these RVs} \\ &= \frac{\prod_i (1 + \mu_i \cdot (e^t - 1))}{e^{t(1+\epsilon)\mu}} && \text{Bernoulli RVs} \\ &\leq \frac{\prod_i (\exp(\mu_i(e^t - 1)))}{e^{t(1+\epsilon)\mu}} \\ &= \big(\exp(\mu(e^t - 1) - t\mu(1+\epsilon))\big)\end{aligned}$$

Now, we will select t to minimze this upper bound, by taking the derivative, setting to zero and solving, which causes us to choose $t = \ln(1+\epsilon)$, which combined with the previous derivation gives

$$\begin{aligned}\mathbb{P}[S_n \geq (1+\epsilon)\mu] &\leq e^{-\mu(1+\epsilon)\ln(1+\epsilon)+\mu\epsilon} \\ &= e^{\mu(-(1+\epsilon)\ln(1+\epsilon)+\epsilon)} \\ &\leq e^{\mu\left(-(1+\epsilon)\frac{\epsilon}{1+\frac{\epsilon}{2}}+\epsilon\right)} && \text{Since } \ln(1+x) \geq \frac{x}{1+\frac{x}{2}} \text{ for } x > 0 \\ &\leq e^{\frac{-\epsilon^2\mu}{2+\epsilon}}\end{aligned}$$

as desired. If one alternatively simplifies the previous upper bound, one can also show

$$\mathbb{P}[S_n \geq (1+\epsilon)\mu] \leq \left(\frac{e^{\epsilon}}{(1+\epsilon)^{(1+\epsilon)}}\right)^{\mu}$$

which can be useful in different contexts.

To remove the assumption of the variables being bernoulli, notice that the only thing we used about them being bernoulli was that $\mathbb{E}[e^{tX}] = (1+\mu(e^t-1))$. For some random variable Y with mean μ which is not Bernoulli, replace Y with a Bernoulli RV with the same mean. Since e^x is convex, we know that $\mathbb{E}[e^{tY}] \leq \mathbb{E}[e^{tX}]$, so the same analysis applies.

The lower tail bound follows from a similar analysis. □

4 Other Useful Concentration Inequalities

Here, we state without proof several other useful concentration inequalities for your research convenience.

Theorem 4 (Bernstein's Inequality [McDiarmid, 1998])**.** *Consider n independent random variables $X_1, X_2, \ldots, X_n$ with $|(X_i - E[X_i]| \leq b$ for each i. Let $S_n = X_1 + X_2 + \ldots + X_n$, and let S_n have mean μ and variance σ^2. Then, for any $\beta \geq 0$, we have that*

$$\textit{Upper tail:} \quad \mathbb{P}[S_n \geq \mu(1+\beta)] \leq \exp\left(-\frac{\beta^2\mu}{2\sigma^2/\mu + 2\beta b/3}\right)$$

Theorem 5 (McDiarmid's inequality [McDiarmid, 1998])**.** *Consider n independent random variables $X_1, X_2, \ldots, X_n$ with X_i taking values in a set A_i for each i. Suppose a real-valued function f is defined on $\prod A_i$ satisfying $|f(x) - f(x')| \leq c_i$ whenever x and x' differ only in the ith coordinate. Let μ be the expected value of the random variable $f(X_1, \ldots, X_n) = f(X)$. Then for any non-negative β we have*

$$\begin{aligned} &\textit{Upper tail:} \quad \Pr[f(X) \geq \mu(1+\beta)] \leq \exp\left(-\frac{2\mu^2\beta^2}{\sum_i c_i^2}\right) \\ &\textit{Lower tail:} \quad \mathbb{P}[f(X) \leq \mu(1-\beta)] \leq \exp\left(-\frac{2\mu^2\beta^2}{\sum_i c_i^2}\right) \end{aligned} \tag{1}$$

Theorem 6 (Philips and Nelson [1995] show moment bounds are tighter than Chernoff-Hoeffding bounds)**.** *Consider n independent random variables $X_1, X_2, \ldots, X_n$, each with mean 0. Let $S_n = \sum_i X_i$. Then*

$$\mathbb{P}[S_n \geq \lambda] \leq \min_{k \geq 0} \frac{E[X^k]}{\lambda^k} \leq \inf_{t \geq 0} \frac{E[e^{tX}]}{e^{t\lambda}} \tag{2}$$

Theorem 7 (Matrix Chernoff bounds [Tropp et al., 2015])**.** *Consider n independent symmetric matrices $X_1, X_2, \ldots, X_n$ of dimension d. Moreover, $X_i \succeq 0$ and $I \succeq X_i$ for each i, i.e. eigenvalues are all between 0 and 1. Let $\mu_{\min} = \lambda_{\min}\left(\sum \mathbb{E}[X_i]\right)$ and $\mu_{\max} = \lambda_{\max}\left(\sum \mathbb{E}[X_i]\right)$. Then, for any $\gamma \geq 0$*

$$\mathbb{P}[\lambda_{\max}(\sum X_i) \geq \mu_{\max} + \gamma] \leq d\exp(-\frac{\gamma^2}{2\mu_{\max} + \gamma}) \tag{3}$$

and for $\gamma \in [0, 1)$

$$\mathbb{P}[\lambda_{\min}(\sum X_i) \leq \mu_{\min} - \gamma] \leq d\exp(-\frac{\gamma^2}{2\mu_{\min}}) \tag{4}$$

Theorem 8 (Hoeffding-Azuma inequality [McDiarmid, 1998])**.** *Let $c_1, c_2, \ldots, c_n$ be n constants, and let $Y_1, Y_2, \ldots, Y_n$ be a martingale difference sequence with $|Y_i| \leq c_i$ for each i. Then for any $t \geq 0$, we have*

$$\mathbb{P}[|\sum_{i=1}^{n} X_i| \geq t] \leq 2\exp(-\frac{t^2}{2\sum_{i=1}^{n} c_i^2}) \tag{5}$$

References

Colin McDiarmid. Concentration. In *Probabilistic methods for algorithmic discrete mathematics*, pages 195–248. Springer, 1998.

Thomas K Philips and Randolph Nelson. The moment bound is tighter than chernoff's bound for positive tail probabilities. *The American Statistician*, 49(2):175–178, 1995.

Joel A Tropp et al. An introduction to matrix concentration inequalities. *Foundations and Trends® in Machine Learning*, 8(1-2):1–230, 2015.